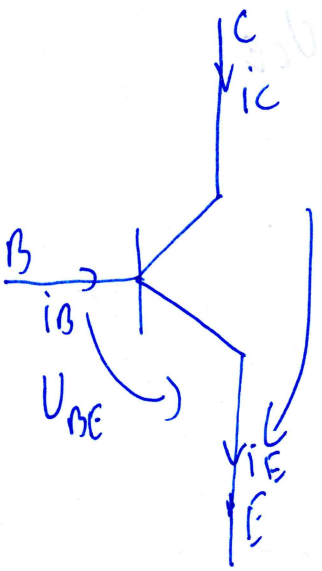
Curs Dela saptămâna 6

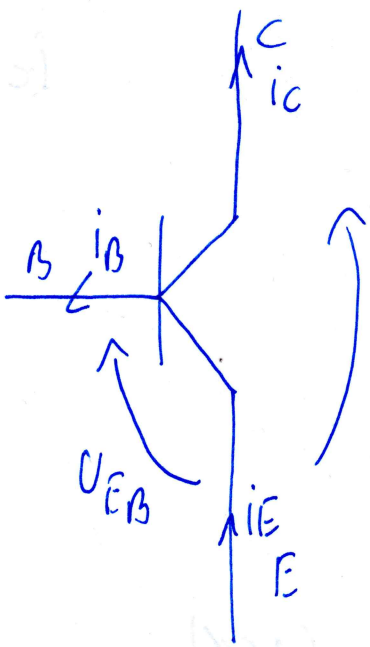
MODEL. SEMNAL MARE

N.P.N.



$U_{BE} = 0,6V$

P.N.P.



$U_{EB} = 0,6V$

RAN:

$i_E = i_C + i_B =$

$= i_C + \frac{i_C}{\beta} \simeq i_C$

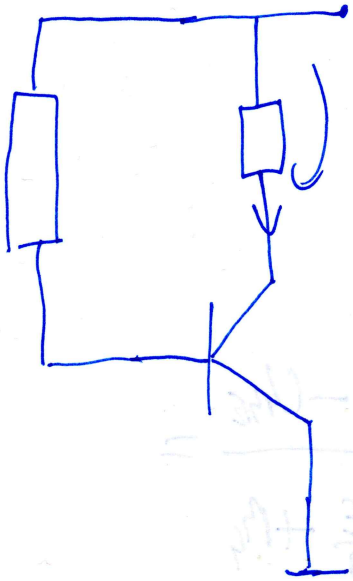
$U_{CE} > U_{BE}$

~~U_{EB}~~ $U_{EC} > U_{EB}$

$i_C = \beta i_B$

$i_C = \beta i_B$

1

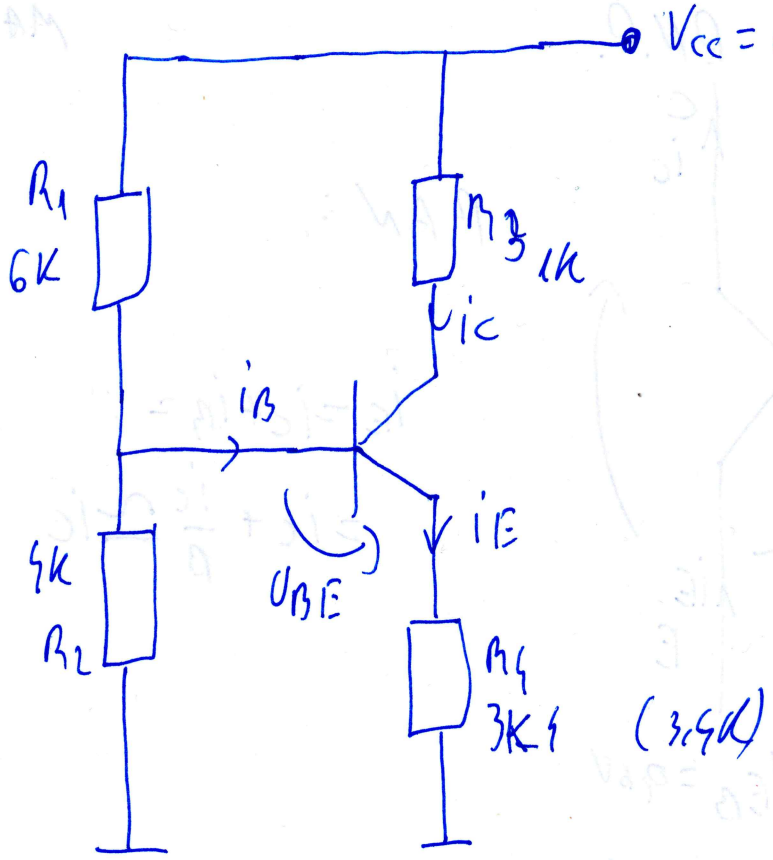


2



Variante a doua mai polarizată

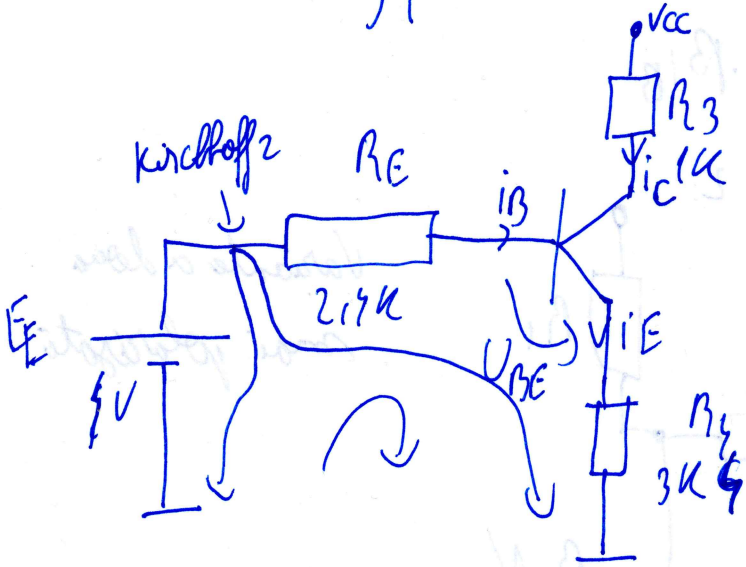
3.



P.S.F

i_C, U_{CE}

S| echivalent (≃)



$$E_E = \frac{R_2}{R_1+R_2} V_{CC} = 4V$$

$$R_E = R_1 \cdot || \, R_2 = 2{,}1K$$

$$E_E = R_E i_B = \cdot U_{BE} + \cdot i_E R_4$$

$$E_E = R_E \cdot \frac{i_E}{\beta} + U_{BE} + i_C \cdot R_4$$

$$\Rightarrow i_C = \frac{E_E - U_{BE}}{\frac{R_E}{\beta} + R_4} =$$

$$= \frac{4V - 0{,}6V}{\frac{2{,}1V}{100} + 3{,}4} = 1mA.$$

$V_{CC} - 0 = R_3 \cdot i_C + U_{CE} + R_4 \cdot i_E$

(potențial de sus - potențial de jos)

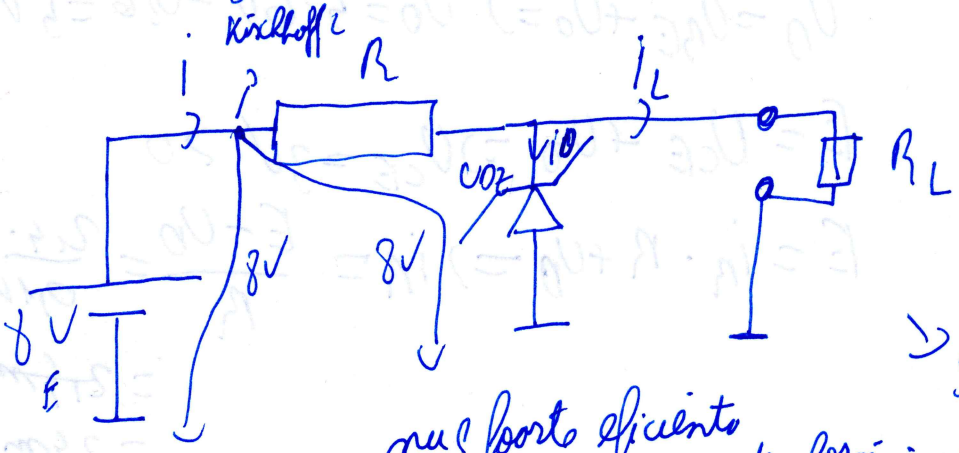
$\Rightarrow U_{CE} = 10V - 1mA \cdot 1k\Omega - 3{,}4V = 5{,}6V > 0{,}6V \Rightarrow$

p.p.că TE în RAN se găsește (regimul activ normal)

dacă $\beta = 200 \Rightarrow I_C = 1mA$

$U_{CE} = 5{,}6V$

STABILIZATOR DE TENSIUNE



→ Stabilizator Paralel

nu e foarte eficient curentul reglorit se transformă în căldură

Ex:

$I_L = 1000mA$

$I_D = 10mA$

$I = I_L + I_D = 1010mA$

$E = RI + U_{DZ}$

$R = \frac{E - U_{DZ}}{I} = \frac{3V}{1000mA} =$

$= 2{,}9\Omega$

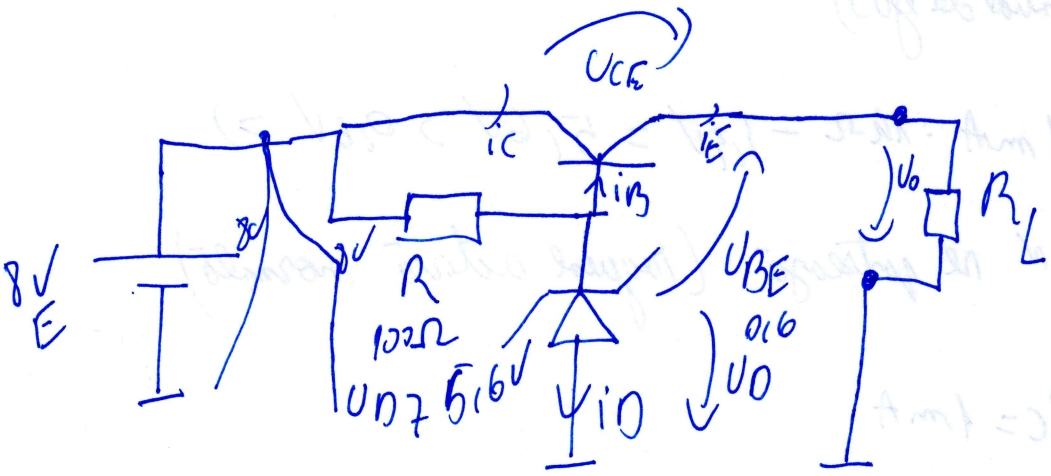
Dacă $I_L = 0$

$I = ?$

$E_E = IR + U_{DZ}$

$I = \frac{E_E - U_{DZ}}{R} = 1010mA$

Modificam schema si adaugam un tranzistor



Dioda - CD $i_D > 0$ $U_0 = U_{D7} = 5{,}6V$

$U_D = U_{BE} + U_0 \Rightarrow U_0 = 5{,}6V - 0{,}6 = 5V$

$E = U_{CE} + U_0 \Rightarrow U_{CE} = 3V > 0$

$E = i_R \cdot R + U_D \Rightarrow i_R = \frac{E - U_D}{R} = \frac{2{,}4}{0{,}1k} =$

$= 24mA$

$i_R = I_B + i_D$

$I_B = \frac{i_C}{\beta} \simeq \frac{I_E}{\beta} = \frac{i_L}{\beta}$

$i_L = 1000mA \Rightarrow i_B = 10mA \Rightarrow i_D = 14mA$

$i_L = 0mA \Rightarrow i_B = 0mA \Rightarrow i_D = 24mA$

a) $i = i_R + i_C \simeq 24mA + 1000mA = 1024mA.$

b) $i = i_R + i_C = 24mA$

j.

	STAB PARALEL	STAB SERIE
$i_L = 1000mA$	1010mA.	1024mA
$i_L = 0mA$	1010mA	24 mA
	consum	consum